Supply Chain Case Study

Data-Driven Recommendations for
Cost-Effective Production and Logistics

Hello, and thank you for the opportunity to present.

In this analysis, I evaluated supplier selection and shipping strategy for Tool Depot’s hammer product launch, focusing on optimizing total cost while ensuring scalability as demand grows.

I’ll walk through the problem context, the analytical approach, key findings, and the final recommendation supported by the data.

Tool Depot is preparing to launch hammers as a new product and must select a single supplier strategy for both locations.

To support this decision, I analyzed supplier pricing, shipping cost structures, carrier capacity constraints, and projected demand growth to evaluate total cost across multiple scenarios.

Based on this analysis, I recommend selecting Supplier B for hammer production and leveraging Carrier X's available capacity to optimize overall logistics efficiency.

As a secondary strategy, I propose piloting a split-shipment program to validate how these cost assumptions and operational conditions behave as order volumes scale.

This approach ensures that the selected strategy is not only cost-effective in the short term, but remains effective as demand increases and supply chain dynamics evolve.

Current Performance & Cost Analysis

- Supplier B has higher production costs, but transportation can offset this
- Carrier X has excess capacity; Carrier Y offers flexible rates
- Shipment weights and costs analyzed for several scenarios

Yearly Carrier X

	Hammer Cost Per Unit	Hammer Product Cost	Transportation Cost Store 1	Total Cost	Transportation Cost Store 2	Total Cost	Total
Supplier A	$ 0.80	$ 722,770	$ 132,080	$ 854,850	$ 85,280	$ 808,050	$ 1,662,901
Supplier B	$ 0.82	$ 740,840	$ 62,400	$ 803,240	$ 62,400	$ 803,240	$ 1,606,479
Diff	$ (0.02)	$ (18,069.26)	$ 69,680.00	$ 51,611	$ 22,880	$ 4,811	$ 56,421

Yearly Carrier Y

	Hammer Cost Per Unit	Hammer Product Cost	Transportation Cost Store 1	Total Cost	Transportation Cost Store 2	Total Cost	Total
Supplier A	$ 0.80	$ 722,770	$ 108,416	$ 831,186	$ 63,242	$ 786,013	$ 1,617,199
Supplier B	$ 0.82	$ 740,840	$ 63,242	$ 804,082	$ 54,208	$ 795,047	$ 1,599,130
Diff	$ (0.02)	$ (18,069.26)	$ 45,173.15	$ 27,104	$ 9,035	$ (9,035)	$ 18,069

If we evaluate hammer production in isolation, Supplier B with Carrier Y initially appears to be the lowest-cost option.

However, when analyzing total landed cost, shipment data shows that Carrier X has unused capacity on existing loads. By assigning Supplier B's hammers to Carrier X, we can utilize this excess capacity, reduce marginal shipping costs, and improve overall logistics efficiency.

This shifts the decision from a unit-cost comparison to a system-level optimization, where transportation efficiency becomes the primary driver of total cost.

Our projections confirm that combined loads remain within carrier capacity limits, while also providing a more scalable and cost-efficient solution as order volumes grow.

Primary Recommendation: Leverage Supplier B with Carrier X

- Standalone, Supplier B + Carrier Y is cheapest for hammers
- Shipment data shows Carrier X's unused capacity can be leveraged
- Pivot: Assign Supplier B's hammers to Carrier X for cost and efficiency gains

Store 1 Shipping Merged below 44,000	Store 2 Shipping Merged below 44,000	Store 1 Shipping with Hammers Merged below 44,000	Store 2 Shipping with Hammers Merged below 44,000
32,646.2	25,967.7	53,642.2	43,223.4
32,828.0	23,694.2	55,082.3	39,562.2
30,206.2	22,617.3	50,863.3	38,887.0
28,824.2	27,455.2	45,536.7	50,307.6
32,471.8	23,979.1	49,557.0	41,746.1
27,554.1	23,646.4	43,468.0	42,317.2
33,469.0	28,927.3	52,824.2	49,880.6
28,283.9	23,700.9	46,975.9	41,184.9
26,201.8	23,694.7	41,970.5	41,041.7
30,755.3	25,617.7	51,218.9	44,224.6
27,867.4	25,657.1	44,952.6	44,373.6
32,974.0	25,011.8	53,432.7	42,267.5
24,740.3	22,602.6	39,211.9	41,647.8
29,733.2	24,827.2	46,639.3	42,776.8
37,630.1	26,193.8	61,046.0	43,312.5

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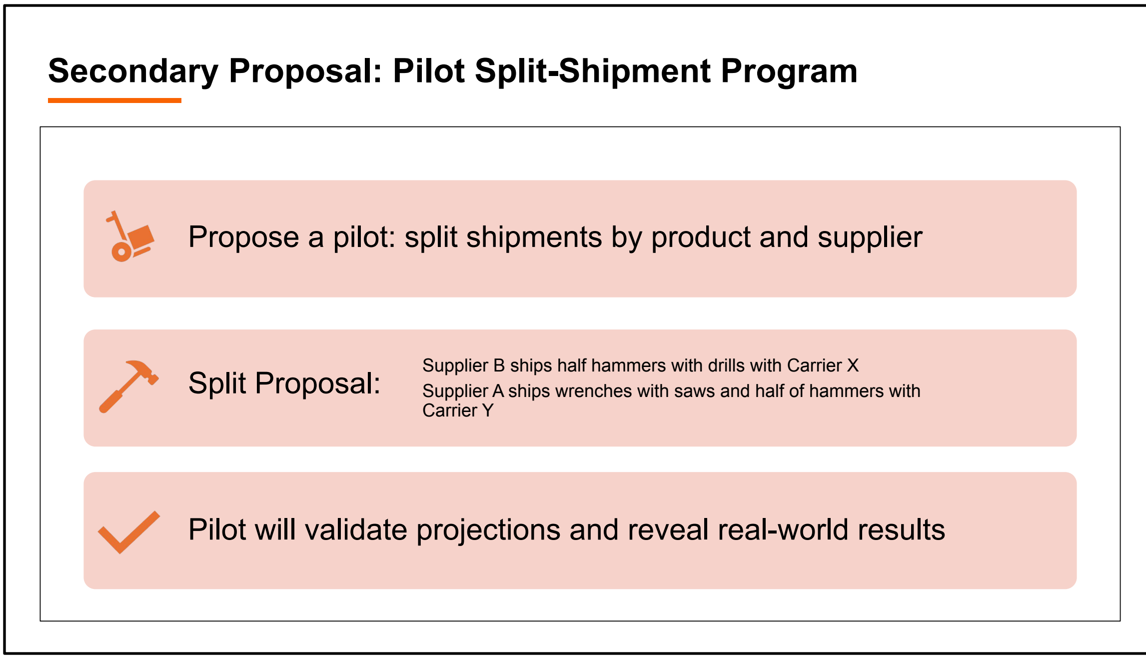
This shifts the focus from unit cost to system-level cost optimization, where transportation efficiency becomes the primary driver of total cost.

Our projections confirm that this combined approach remains within carrier capacity limits while delivering a more efficient and scalable logistics solution.



Looking at the shipment and cost data over a multi-year period, we see a clear pattern: as order volumes grow, shipment weights approach the carrier’s capacity limit. By year five, we reach that threshold, which forces a shift in shipping strategy to maintain cost efficiency.

Piloting a split lineup using both suppliers for their respective products allows us to proactively manage this transition. It reduces the risk of production slowdowns and helps us avoid disruptive carrier changes as volumes increase.

This pilot is not just about testing logistics; it is about validating how our cost assumptions behave as the system scales. It allows us to observe how carrier pricing structures respond to increasing volume and identify any changes in cost efficiency over time.

This approach also provides flexibility in supplier selection, reducing reliance on a single sourcing strategy and ensuring continuity if cost or capacity conditions shift.

Ultimately, the pilot positions Tool Depot to make a long-term decision based on real operational data rather than initial projections, ensuring the chosen strategy remains effective under sustained growth.

Why This Approach? Data-Backed Rationale

- Projections and Historical Data Support
- The Rationale: Our analysis of projections and historical data confirms the viability of both proposals.
- Split Approach Aligns with Core Metrics
- The Rationale: The recommended split approach is specifically designed to align with existing load limits while optimizing cost trends.
- Ongoing Monitoring Ensures Adaptability
- The Rationale: We will implement ongoing monitoring to ensure adaptability in the face of new challenges and robust risk management.

The pilot program allows us to collect real-world data on costs, capacity utilization, and operational performance.

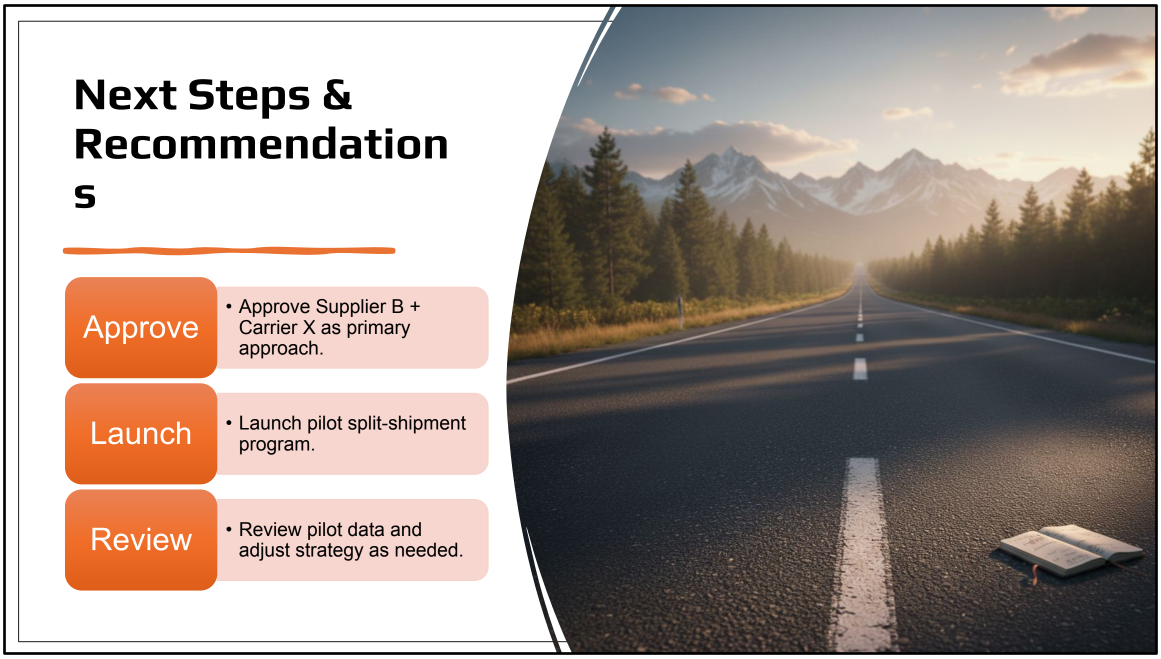
By monitoring how shipments perform as we approach carrier weight limits, we can confirm whether projected cost savings and efficiency gains hold as volume increases.

One key risk identified in the analysis is that the lowest-cost option in the short term may not remain the most cost-effective over time.

As demand grows, variable shipping costs can begin to scale faster than flat-rate capacity, creating a cost inflection point where initial savings diminish.

This means a decision based solely on upfront cost could lead to higher total cost over the life of a long-term contract.

The pilot program enables us to validate this behavior in real-world conditions, ensuring the selected strategy remains financially and operationally effective as the system scales.



Based on the analysis, I recommend moving forward with Supplier B and Carrier X for hammer production and shipping, leveraging their strengths for immediate efficiency.

At the same time, we should launch the pilot split-shipment program to validate our strategy and gather actionable insights as order volumes grow.

After the pilot, we'll review the results, assess any operational challenges, and adjust our approach as needed to ensure we maintain cost efficiency and supply chain resilience.

This approach also ensures we are not locking into a strategy based solely on short-term cost advantages, but instead validating that it remains effective as demand scales and shipping conditions evolve.

This dual-track strategy positions Tool Depot to scale effectively while minimizing risk and maximizing long-term value.

Q&A

That concludes my analysis and recommendations for Tool Depot's hammer supplier and shipping strategy.

I'm happy to answer any questions, clarify any part of the approach, or discuss how these recommendations would perform under different growth or operational scenarios.

I'd also welcome the opportunity to explore how this strategy could be refined further as additional data becomes available.

Thank you for your time, and I look forward to your feedback.